

stat_rosenbaum_lf_family_minimax

Specialization: stat

TL;DR. Conditioning on the observed records turns the Rosenbaum sharp-null problem into an exact finite all-tests linear program. The Rosenbaum-specific finite contribution is the product-vertex program, the indexed-support criterion $D_{\mathcal{B}} = D_{\mathcal{V}}$, the exact classical-validity topology, and auditable support-switch certificates. In the three-position example, $\mathcal{A}_1 = (0, 2/3]$; at level $7/10$, the optimal rule is $(1/10, 7/10, 1)$, its power is $197/260$, and the least-favorable weights are $16/21$ and $5/21$. A separate 84-vertex diagnostic proves that coordinatewise monotonicity can cost exactly $1/720$. In the fixed-record, fixed-score replicated oracle experiment, with positions 1 and m labeling the score minimum and maximum, the all-randomized-tests design-sensitivity boundary is $\tilde{\Gamma}_{\text{all}}^*(Q) = Q(m)/Q(1)$. Below it, the explicit endpoint score $s_{\Gamma}(j) = \mathbf{1}\{j = m\} - \Gamma \mathbf{1}\{j = 1\}$ yields uniform null size and alternative-error bounds $e^{-c_0 n}$, where $c_0 = \Delta^2/[2(\Gamma + 1)^2]$; at and above it, the frontier equals α for every n . For $m = 2$, the boundary is $Q(2)/Q(1) = q_2/(1 - q_2)$, the sign/McNemar success odds.

1 Project justification

Gap. Generic finite minimax theory does not expose the Rosenbaum product vertices, decide when a prescribed indexed terminal-block support is sufficient, identify the levels where the classical tilt is minimax sharp, or provide exact support-switch certificates. Existing asymptotic sensitivity results study selected procedures or sensitivity-value estimators rather than the maximal common- Γ detectability boundary over every randomized test.

Niche. The note joins two sharply delimited objects: an exact conditional finite all-tests linear program for heterogeneous matched sets and an oracle repeated-design benchmark with fixed m , fixed record, fixed endpoint-extremal Q , common Γ , and independent replicas. The causal target remains the Fisher sharp null; the finite frontier and replicated design-sensitivity boundary are testing functionals, not alternative definitions of the causal null.

Fill. Vertex reduction makes composite-null validity finitely auditable. Linear-program duality and complementary slackness give the least-favorable mixture, assignment-dependent likelihood-ratio ties, the support equality $D_{\mathcal{B}} = D_{\mathcal{V}}$, the tie-block representation of $\mathcal{A}_n$, and the affine support-switch envelope. In the replicated design, the explicit endpoint score $s_{\Gamma}(j) = \mathbf{1}\{j = m\} - \Gamma \mathbf{1}\{j = 1\}$ has null supremum zero and alternative mean $\Delta = [Q(m) - \Gamma Q(1)]/\sum_k Q(k)$. Hoeffding's inequality then gives both errors at most $e^{-c_0 n}$, while $q \in \mathcal{P}_m(\Gamma)$ at and above the boundary gives the exact all-tests converse.

2 Related work

Rosenbaum's multiple-control analysis gives sharp sensitivity calculations for a selected generalized signed-rank family [2]. Gastwirth–Krieger–Rosenbaum establish asymptotic separability for specified rank procedures with multiple controls [4]; their result concerns selected procedures, whereas the replicated theorem here is an oracle detectability boundary over all randomized tests. Zhao derives asymptotic normality for the pair-matched sensitivity value [5]; by contrast, $q_2/(1 - q_2)$ here is a population all-tests boundary, not an estimator or a sampling distribution. Cohen–Olson–Fogarty study coherent combinations of prespecified outcome statistics [6]; the present benchmark ranges over all randomized tests but is oracle and is not a practical coherent-statistic construction. Fogarty studies limiting laws and design sensitivity for tilted candidate statistics [8]; the present result instead identifies the maximal common- Γ all-tests boundary and makes no operational-domination claim. Fogarty–Small share hidden-assignment probabilities across multiple outcomes [7], a different optimization from the present single-outcome conditional frontier. Huber–Strassen provide generic robust-testing background, but their representative-pair conclusion uses a capacity structure not established for this Rosenbaum product-law envelope [3]. The contribution here is the Rosenbaum-specific finite program and certificates together with the explicit endpoint-score repeated-design boundary, its uniform Hoeffding achievability, and its exact all-tests converse.

3 Setup

Target (estimand / structural parameter): $H_{\{F, n\}}: Y_{\{ij\}}(1) = Y_{\{ij\}}(0) \setminus \text{forall}(i, j)$. Identifying functional / recovery map: $\beta_n^*(\alpha; Q, \Gamma, r) = \sup_{\{\phi: \sup_{\{P \in \mathcal{M}_{\{0, n\}}(\Gamma; r)\}} \mathbb{E}_{\phi} [Y_{\{ij\}}(1) - Y_{\{ij\}}(0)] \leq \alpha\}} \mathbb{E}_{\phi} [Y_{\{ij\}}(1) - Y_{\{ij\}}(0)]$.

$P_{\phi|e|\alpha} E_{\{P_{1,n}^Q(\cdot \mid r)\}|\phi}$.

- n — positive integer; $\mathbb{N}$; array index

Definition. number of matched sets

- i — index; $\{1, \dots, n\}$; index

Definition. matched-set index

- m_i — integer; $\{2, 3, \dots\}$; set size

Definition. number of positions in matched set i

- j — index; $\{1, \dots, m_i\}$; index

Definition. position index in matched set i

- a — treatment label; $\{0, 1\}$; causal index

Definition. potential-treatment label

- $Y_{ij}(a)$ — random potential outcome; $\mathbb{R}$; causal primitive

Definition. potential outcome for position j in set i under treatment a

- $H_{F,n}$ — causal hypothesis; $\{0, 1\}$; causal target

Definition. Fisher sharp null $Y_{ij}(1) = Y_{ij}(0)$ for every i, j

- Γ_i — scalar; $[1, \infty)$; sensitivity parameter

Definition. Rosenbaum hidden-bias bound for set i

- α — scalar; $(0, 1)$; level parameter

Definition. conditional test level

- Ω_n — finite set; $\prod_{i=1}^n \{1, \dots, m_i\}$; assignment sample space

Definition. assignment vectors with one treated position per set

- Z_i — random variable; $\{1, \dots, m_i\}$; assignment

Definition. treated position in matched set i

- Z — random vector; Ω_n ; assignment vector

Definition. vector of treated positions

- z — assignment realization; Ω_n ; assignment value

Definition. realized vector of treated positions

- $\mathcal{R}_i$ — measurable space; outcome-record space

Definition. sample space for the observed outcome record in matched set i

- R_i — outcome record; $\mathcal{R}_i$; observed causal data

Definition. observed outcome record for matched set i

- r_i — outcome-record realization; $\mathcal{R}_i$; conditioning value

Definition. realized observed outcome record in matched set i

- r — outcome-record realization; $\mathcal{R}_1 \times \dots \times \mathcal{R}_n$; conditioning value

Definition. full observed outcome record $r = (r_1, \dots, r_n)$

- $F_{\{0i\}}$ — probability law; $\mathcal{P}(\mathcal{R}_i)$; sharp-null outcome law
Definition. law of R_i under $H_{F,n}$, which does not depend on the treated position
- $F_{\{1ij\}}$ — probability law; $\mathcal{P}(\mathcal{R}_i)$; causal alternative law
Definition. law of R_i under a specified causal potential-outcome alternative when position j is treated
- $\mathcal{R}_i^Q$ — measurable set; $\mathcal{R}_i$; likelihood-ratio support
Definition. common record set on which every selected alternative likelihood ratio is finite and positive
- $Q_i(j; r_i)$ — positive scalar; $(0, \infty)$; conditional alternative likelihood score
Definition. selected Bayes-factor density $dF_{1ij}/dF_{0i}(r_i)$ on $\mathcal{R}_i^Q$
- Q — score array; $\prod_{i=1}^n (0, \infty)^{m_i}$; alternative input
Definition. collection of conditional causal Bayes-factor scores at r
- π_i — probability vector; $\Delta(\{1, \dots, m_i\})$; alternative assignment prior
Definition. specified uniform alternative assignment baseline in set i
- $P_{\{\mathrm{alt}, n\}}$ — probability law; $\mathcal{P}(\Omega_n \times \mathcal{R}_1 \times \dots \times \mathcal{R}_n)$; alternative data-generating law
Definition. joint law of assignment and outcome record under the specified causal alternative
- P — probability law; $\mathcal{P}(\Omega_n)$; null-law candidate
Definition. conditional assignment law given r under the sharp null
- $\mathcal{M}_{\{0,n\}}(\Gamma; r)$ — class of laws; $\mathcal{P}(\Omega_n)$; composite null
Definition. conditional Rosenbaum null assignment-law class
- $P_{\{1,n\}}^Q(\cdot \mid r)$ — probability law; $\mathcal{P}(\Omega_n)$; conditional power alternative
Definition. posterior conditional assignment law under $P_{\mathrm{alt},n}$ after conditioning on $R = r$
- ϕ — conditional test; $[0,1]^{\Omega_n}$; decision rule
Definition. possibly randomized rejection rule after conditioning on r
- $\beta_n^*(\alpha; Q, \Gamma, r)$ — scalar; $[0,1]$; power functional
Definition. conditional all-tests power frontier against the causal alternative
- S_i — set; $2^{\{1, \dots, m_i\}}$; vertex index
Definition. high-probability positions of a local Rosenbaum-polytope vertex
- $v_{\{i, S_i\}}$ — probability vector; $\Delta(\{1, \dots, m_i\})$; local vertex
Definition. local extreme-tilt assignment vector indexed by S_i
- P_v — probability law; $\mathcal{P}(\Omega_n)$; global null vertex
Definition. global product law generated by local extreme tilts
- $\mathcal{V}_n$ — finite set of laws; $\mathcal{P}(\Omega_n)$; finite null reduction
Definition. product extreme laws of the conditional Rosenbaum null polytope

- $\mathcal{V}_n^Q(r)$ — finite set of laws; $\mathcal{V}_n$; indexed support candidate
Definition. indexed terminal-block candidate vertices; under the additional strict score-ordering assumption, these terminal blocks are upper sets of the conditional Bayes-factor ordering
- $\mathbf{p}_i^+$ — probability vector; $\Delta(\{1, \dots, m_i\})$; classical local null
Definition. local classical tilt placing high probability on position m_i
- $\mathbf{P}_n^+$ — probability law; $\mathcal{P}(\Omega_n)$; candidate representative null
Definition. product of the local classical extreme tilts
- $\mathbf{P}_0$ — probability law; $\mathcal{P}(\Omega_n)$; testing input
Definition. candidate simple null law in a Neyman--Pearson rule
- c — scalar; $[0, \infty)$; test cutoff
Definition. likelihood-ratio cutoff
- ρ — boundary randomization function; $[0, 1]^{\Omega_n}$; test randomization
Definition. assignment-dependent rejection probability on a likelihood-ratio tie set
- λ — probability vector; $\Delta(\mathcal{V}_n)$; least-favorable convexification parameter
Definition. mixing weights over global Rosenbaum vertices; support may optionally be restricted to $\mathcal{V}_n^Q(r)$
- $\mathbf{P}_\lambda$ — probability law; $\operatorname{conv}(\mathcal{V}_n) \subseteq \mathcal{P}(\Omega_n)$; convexified least-favorable adversary
Definition. convex mixture of global vertex laws; unless degenerate, it need not be conditionally independent or belong to $\mathcal{M}_{0,n}(\Gamma; r)$
- $\mathcal{A}_n(Q, \Gamma; r)$ — set; $(0, 1)$; Rosenbaum validity region
Definition. levels at which at least one generalized Neyman--Pearson rule calibrated under the classical tilt is valid for the full conditional Rosenbaum null

4 Assumptions

Assumption 1 (ass:sharp-null). *The Fisher sharp null holds: $Y_{ij}(1) = Y_{ij}(0)$ for every i and j .*

Standard (*Fisher sharp null*, [1]).

Assumption 2 (ass:conditional-independence). *Conditional on r under the sharp null, $Z_1, \dots, Z_n$ are independent.*

Standard (*independent matched-set assignments*, [1]).

Assumption 3 (ass:rosenbaum-odds). *For every i and $j, k \in \{1, \dots, m_i\}$, $\Pr_P(Z_i = j \mid r) / \Pr_P(Z_i = k \mid r) \in [\Gamma_i^{-1}, \Gamma_i]$.*

Standard (*Rosenbaum bounded-odds sensitivity model*, [1]).

Assumption 4 (ass:alternative-prior-factorization). *Under $P_{\text{alt},n}$, $\Pr(Z = z) = \prod_{i=1}^n \pi_i(z_i)$ for every $z \in \Omega_n$.*

Novel. *This is an explicit alternative-model premise, not a consequence of the Rosenbaum null. It describes a prospectively randomized causal alternative in which treatment is assigned uniformly within independently randomized matched sets before the hidden-bias sensitivity model is imposed for the null comparison.*

Assumption 5 (ass:alternative-record-factorization). *Under $P_{\text{alt},n}$, $\Pr(R \in dr \mid Z = z) = \prod_{i=1}^n F_{1iz_i}(dr_i)$ for every $z \in \Omega_n$.*

Novel. *This is an explicit alternative-model premise, not a property of pair-matched sensitivity-value theory. It is realistic for independently recruited matched sets with no cross-set interference or shared post-match shocks; for example, separately sampled matched strata with unit-level outcomes generated after one within-set treatment assignment.*

Assumption 6 (ass:likelihood-common-support). For every i, j , F_{1ij} and F_{0i} are equivalent measures, and dF_{1ij}/dF_{0i} has a finite strictly positive F_{0i} -almost-everywhere measurable version.

Standard (common support for likelihood-ratio testing, [3]).

Assumption 7 (ass:conditioning-support). For every i , the conditioning record satisfies $r_i \in \mathcal{R}_i^Q$.

Standard (common-support conditioning, [3]).

Assumption 8 (ass:causal-likelihood-order). For every i , the causal Bayes factors obey $0 < Q_i(1; r_i) < \dots < Q_i(m_i; r_i)$.

Novel. The condition holds when a prespecified ordered causal mechanism makes the observed outcome record progressively more likely as the treated position moves up an ordered within-set response contrast. For example, it is satisfied by monotone exponential-family outcome alternatives with a strictly ordered sufficient-statistic contrast.

Assumption 9 (ass:causal-likelihood-separation). For every i , $Q_i(m_i; r_i)/Q_i(1; r_i) > \Gamma_i$.

Novel. This is a finite conditional signal-strength condition: the causal outcome evidence distinguishing the two extreme treated positions exceeds the allowed hidden-assignment odds. It is met in ordered binary or exponential-family alternatives once the treatment-induced log-likelihood contrast is larger than $\log \Gamma_i$.

Assumption 10 (ass:test-level). The conditional level satisfies $0 < \alpha < 1$.

Standard (nondegenerate test level, [3]).

Assumption 11 (ass:replicated-conditional-design). Fix an integer $m \geq 2$, a conditioning record r_0 , a common $\Gamma \geq 1$, and a score vector $Q = (Q(1), \dots, Q(m)) \in (0, \infty)^m$ whose endpoint labels are extremal, $Q(1) = \min_j Q(j) < \max_j Q(j) = Q(m)$, and require for every n that the matched sets are conditionally independent replicas with $m_i = m$, $r_i = r_0$, $\Gamma_i = \Gamma$, conditional alternative $q^{\otimes n}$, where $q_j = Q(j)/\sum_{k=1}^m Q(k)$, and conditional sharp-null class exactly $\{\bigotimes_{i=1}^n p_i : p_i \in \mathcal{P}_m(\Gamma) \text{ for every } i\}$.

Novel. novel: this is an explicit fixed-record, fixed-score conditional oracle experiment used to define the repeated-design benchmark; it is not the theorem's conclusion.

5 Definitions

Definition 1 (def:null-outcome-law). Defined object. *sharp-null outcome law*

Construction. $F_{\{0i\}} = \mathcal{L}(R_i \mid H_{\{F, n\}})$

Inputs. $R_i; H_{\{F, n\}}; i$.

Definition 2 (def:alternative-prior). Defined object. *uniform alternative assignment baseline*

Construction. $p_{i \cdot i}(j) = m_i^{-1} \text{ for } j \in \{1, \dots, m_i\}$

Inputs. $p_{i \cdot i}; j; m_i$.

Definition 3 (def:common-record-support). Defined object. *common likelihood-ratio record set*

Construction. $\mathcal{R}_i^Q = \{s \in \mathcal{R}_i : 0 < dF_{\{1ij\}}/dF_{\{0i\}}(s) < \infty \text{ for every } j \in \{1, \dots, m_i\}\}$

Inputs. $F_{\{1ij\}}; F_{\{0i\}}; \mathcal{R}_i; i; j; m_i$.

Definition 4 (def:causal-bayes-factor). Defined object. *causal conditional Bayes factor*

Construction. $Q_i(j; s) = dF_{\{1ij\}}/dF_{\{0i\}}(s) \text{ for } s \in \mathcal{R}_i^Q$

Inputs. $F_{\{1ij\}}; F_{\{0i\}}; \mathcal{R}_i^Q; i; j$.

Definition 5 (def:gamma-null). Defined object. *conditional Rosenbaum null class*

Construction. $\mathcal{M}_{\{0, n\}}(\Gamma; r) = \{P \in \mathcal{P}(\Omega_n) : P \text{ satisfies } \text{ass:conditional-independence} \text{ and } \text{ass:rosenbaum-odds}\}$

Member properties. *ass:conditional-independence; ass:rosenbaum-odds.*

Definition 6 (def:alternative-law). Defined object. *conditional causal alternative law*

Construction. $P_{\{1,n\}} \sim Q(z \mid r) = \Pr_{\{P_{\{\mathrm{alt}\},n\}}(Z=z \mid R=r) = \prod_{i=1}^n \frac{\pi_i(z_i)Q_i(z_i;r_i)}{\sum_{k=1}^m \pi_i(k)Q_i(k;r_i)} = \prod_{i=1}^n \frac{Q_i(z_i;r_i)}{\sum_{k=1}^m Q_i(k;r_i)} \quad \text{for } z \in \Omega_n$

Inputs. $P_{\{\mathrm{alt}\},n}; \pi_i; Q_i(j;r_i); r; n; m_i; Z; \text{ass:alternative-prior-factorization}; \text{ass:alternative-record-factorization}; \text{def:alternative-prior}.$

Definition 7 (def:frontier). Defined object. *conditional all-tests causal power frontier*

Construction. $\beta_n^*(\alpha; Q, \Gamma, r) = \sup_{\{\phi \in [0,1] \mid \mathbb{E}_{P_{\{1,n\}} \sim Q}(\phi(Z)) \geq \alpha\}} \mathbb{E}_{P_{\{1,n\}} \sim Q}(\phi(Z))$

Inputs. $\phi; P_{\{1,n\}} \sim Q; \Gamma; r; \alpha.$

Definition 8 (def:extreme-laws). Defined object. *global conditional Rosenbaum extreme laws*

Construction. $v_{\{i,S\}}(j) = \Gamma_i / (|S| \Gamma_{i+m_i-|S|}) \text{ if } j \in S \text{ and } v_{\{i,S\}}(j) = 1 / (|S| \Gamma_{i+m_i-|S|}) \text{ if } j \notin S; \quad \mathcal{V}_n = \{P_v : P_v(z) = \prod_{i=1}^n v_{\{i,S_i\}}(z_i), \varnothing \subsetneq S_i \subsetneq \{1, \dots, m_i\} \text{ for every } i\}$

Inputs. $\Gamma_i; m_i; n; Z.$

Definition 9 (def:monotone-vertices). Defined object. *conditional indexed upper-set-candidate extreme laws*

Construction. $\mathcal{V}_n^Q(r) = \{P_v \in \mathcal{V}_n : S_i = \{m_i - k_i + 1, \dots, m_i\} \text{ for some } k_i \in \{1, \dots, m_i - 1\} \text{ for every } i\}$

Inputs. $\mathcal{V}_n; Q_i(j;r_i); m_i; n; r.$

Definition 10 (def:classical-tilt). Defined object. *classical conditional extreme-tilt law*

Construction. $P_{n^+}(z) = \prod_{i=1}^n p_i^+(z_i), \quad p_i^+(m_i) = \Gamma_i / (\Gamma_{i+m_i-1}), \quad p_i^+(j) = 1 / (\Gamma_{i+m_i-1}) \text{ for } j < m_i$

Inputs. $\Gamma_i; m_i; n; Z.$

Definition 11 (def:np-rule). Defined object. *generalized conditional Neyman--Pearson rule*

Construction. $\phi_{\{\alpha, P_0\}} \sim \{\mathrm{NP}\}(z; r) = \mathbb{1}_{\{P_{\{1,n\}} \sim Q(z \mid r) > cP_0(z)\}} + \rho(z) \mathbb{1}_{\{P_{\{1,n\}} \sim Q(z \mid r) = cP_0(z)\}}, \quad c \geq 0, \rho \in [0,1] \sim \Omega_n, \quad \mathbb{E}_{P_0}[\phi_{\{\alpha, P_0\}} \sim \{\mathrm{NP}\}(Z; r)] = \alpha; \text{ the vector } \rho \text{ is assignment-dependent on the tie set and is selected jointly with } c \text{ to meet every additional composite-null feasibility and complementary-slackness condition imposed by the testing problem}$

Inputs. $P_{\{1,n\}} \sim Q; P_0; \alpha; c; \rho; Z; r.$

Definition 12 (def:mixture-law). Defined object. *convexified conditional adversary law*

Construction. $P_{\lambda} = \sum_{P_v \in \mathcal{V}_n} \lambda(P_v) P_v, \quad \lambda \in \Delta(\mathcal{V}_n)$

Inputs. $\lambda; \mathcal{V}_n; P_v.$

Definition 13 (def:lf-mixture-handle). Defined object. *least-favorable-mixture finite dual*

Construction. $\min_{\{c \geq 0, \lambda \in \Delta(\mathcal{V}_n)\}} \left\{ c\alpha + \sum_{z \in \Omega_n} [P_{\{1,n\}} \sim Q(z \mid r) - cP_{\lambda}(z)]_+ \right\}; \quad \phi(z) = \mathbb{1}_{\{P_{\{1,n\}} \sim Q(z \mid r) > cP_{\lambda}(z)\}} + \rho(z) \mathbb{1}_{\{P_{\{1,n\}} \sim Q(z \mid r) = cP_{\lambda}(z)\}}, \text{ where the assignment-dependent tie vector is selected so that } \mathbb{E}_{P_{\lambda}}[\phi] = \alpha, \mathbb{E}_{P_v}[\phi] \leq \alpha \quad (P_v \in \mathcal{V}_n), \text{ and } \lambda(P_v)(\alpha - \mathbb{E}_{P_v}[\phi]) = 0 \quad (P_v \in \mathcal{V}_n) \text{ together with the coordinatewise complementary-slackness equations}$

Inputs. $P_{\lambda}; \lambda; \mathcal{V}_n; P_v; \alpha; P_{\{1,n\}} \sim Q; c; \rho; \phi.$

Definition 14 (def:validity-region). Defined object. *conditional Rosenbaum validity region*

Construction. $\backslash\mathrm{mathcal{A}}_n(Q, \backslash\Gamma; r) = \{ \backslash\alpha \in (0, 1) : \text{there exist } c \geq 0 \text{ and assignment-dependent } \backslash\rho \in [0, 1] \text{ such that } \backslash\mathrm{mathbb{E}}_{\{P_n^+\}}[\backslash\phi_{\{\backslash\alpha, P_n^+\}} \backslash\mathrm{mathrm{NP}}\}(Z; r)] = \backslash\alpha \text{ and } \backslash\max_{\{P_v \in \mathrm{mathcal{V}}_n\}} \backslash\mathrm{mathbb{E}}_{\{P_v\}}[\backslash\phi_{\{\backslash\alpha, P_n^+\}} \backslash\mathrm{mathrm{NP}}\}(Z; r)] \leq \backslash\alpha \}$

Inputs. $\backslash\mathrm{mathcal{V}}_n; P_v; P_n^+; \backslash\alpha; \backslash\phi_{\{\backslash\alpha, P_n^+\}} \backslash\mathrm{mathrm{NP}}\}; c; \backslash\rho; r$.

Definition 15 (def:local-rosenbaum-polytope). Defined object. *local Rosenbaum probability polytope*

Construction. $\backslash\mathrm{mathcal{P}}_m(\backslash\Gamma) = \{ p \in [0, 1]^m : \sum_{j=1}^m p_j = 1 \text{ and } p_j \leq \backslash\Gamma p_k \text{ for every } j, k \in \{1, \dots, m\} \}$

Inputs. $\backslash\Gamma; m$.

Definition 16 (def:replicated-frontier). Defined object. *replicated conditional all-tests frontier*

Construction. $\backslash\beta_{\{n, \backslash\mathrm{mathrm{rep}}\}} \backslash\star(\backslash\alpha; Q, \backslash\Gamma) = \sup_{\{\backslash\phi : \{1, \dots, m\} \rightarrow [0, 1]\}} \left\{ \backslash\mathrm{mathbb{E}}_{\{q \otimes n\}}[\backslash\phi(Z)] : \sup_{\{p_1, \dots, p_n \in \backslash\mathrm{mathcal{P}}_m(\backslash\Gamma)\}} \backslash\mathrm{mathbb{E}}_{\{ \otimes_{i=1}^n p_i \}}[\backslash\phi(Z)] \leq \backslash\alpha \right\}; \text{this is the specialization of } \backslash\mathrm{def:frontier} \text{ at } r(n) = (r_0, \dots, r_0) \text{ under } \backslash\mathrm{ass:replicated-conditional-design}, \text{ not a competing estimand}$

Inputs. $n; \backslash\alpha; Q; \backslash\Gamma; \backslash\phi; Z; \mathrm{ass:replicated-conditional-design}; \mathrm{def:local-rosenbaum-polytope}; \mathrm{def:frontier}$.

Definition 17 (def:all-tests-design-sensitivity). Defined object. *all-tests replicated design-sensitivity boundary*

Construction. $\backslash\widetilde{\backslash\Gamma}_{\{\backslash\mathrm{mathrm{all}}\}} \backslash\star(Q) = \sup \left\{ \backslash\Gamma : \lim_{n \rightarrow \infty} \backslash\beta_{\{n, \backslash\mathrm{mathrm{rep}}\}} \backslash\star(\backslash\alpha; Q, \backslash\Gamma) = 1 \right\}$

Inputs. $n; \backslash\alpha; Q; \backslash\Gamma; \mathrm{ass:replicated-conditional-design}; \mathrm{ass:test-level}; \mathrm{def:replicated-frontier}$.

6 Main results

Lemma 1 (lem:vertex-reduction). *Under $\mathrm{ass} : \text{conditional} - \text{independence}$ and $\mathrm{ass} : \text{rosenbaum} - \text{odds}$, every conditional test ϕ satisfies*

$$\sup_{P \in \mathcal{M}_{0,n}(\Gamma; r)} \mathbb{E}_P[\phi(Z)] = \max_{P_v \in \mathcal{V}_n} \mathbb{E}_{P_v}[\phi(Z)]$$

Proof. For each matched set i , let $\mathcal{P}_i$ be the local polytope of vectors $p = (p_1, \dots, p_{m_i})$ satisfying $p_j \geq 0$, $\sum_j p_j = 1$, and $p_j \leq \Gamma_i p_k$ for every j, k . The ratio condition in $\mathrm{ass} : \text{rosenbaum} - \text{odds}$ places the conditional law of Z_i in $\mathcal{P}_i$. Conversely, every member of $\mathcal{P}_i$ is strictly positive: if $p_k = 0$, then $p_j \leq \Gamma_i p_k = 0$ for every j , contradicting normalization. It therefore satisfies the ratio condition.

We first determine the extreme points of $\mathcal{P}_i$. If $\Gamma_i = 1$, all coordinates are equal, so $\mathcal{P}_i$ is the singleton uniform law; this is the value of every vector $v_{i,S}$ in $\mathrm{def:extreme-laws}$. Suppose $\Gamma_i > 1$, let p be extreme, and put $a = \min_j p_j$ and $b = \max_j p_j$. If $b < \Gamma_i a$, every ratio inequality is strict. Since $m_i \geq 2$, for distinct j, k and all sufficiently small $\varepsilon > 0$, both $p + \varepsilon(e_j - e_k)$ and $p - \varepsilon(e_j - e_k)$ remain in $\mathcal{P}_i$, contradicting extremality. Hence $b = \Gamma_i a$.

Let $A = \{j : p_j = a\}$, $B = \{j : p_j = b\}$, and $C = \{1, \dots, m_i\} \setminus (A \cup B)$. The active ratio equalities are precisely $p_j = \Gamma_i p_k$ with $j \in B$ and $k \in A$. If $C \neq \emptyset$, fix $c_0 \in C$ and define a nonzero vector d by $d_j = t$ on A , $d_j = \Gamma_i t$ on B , $d_{c_0} = -(|A| + \Gamma_i |B|)t$, and $d_j = 0$ on $C \setminus \{c_0\}$, where $t \neq 0$. Then $\sum_j d_j = 0$, and d preserves every active equality. Finiteness of the inactive strict inequalities and positivity of p imply that, for sufficiently small $\varepsilon > 0$, both $p + \varepsilon d$ and $p - \varepsilon d$ belong to $\mathcal{P}_i$, again contradicting extremality. Thus $C = \emptyset$. Since $\Gamma_i > 1$, both A and B are nonempty, and normalization gives

$$p_j = \frac{\Gamma_i}{|B|\Gamma_i + m_i - |B|} \quad (j \in B), \quad p_j = \frac{1}{|B|\Gamma_i + m_i - |B|} \quad (j \notin B).$$

Therefore $p = v_{i,B}$. Conversely, at $v_{i,S}$, the active equalities $p_j = \Gamma_i p_k$ for $j \in S$ and $k \notin S$, together with $\sum_j p_j = 1$, define a singleton face. Hence $v_{i,S}$ is extreme. The local extreme points are exactly the two-level tilts in $\text{def} : \text{extreme} - \text{laws}$, with harmless duplication when $\Gamma_i = 1$.

By $\text{ass} : \text{conditional} - \text{independence}$ and $\text{def} : \text{gamma} - \text{null}$, each $P \in \mathcal{M}_{0,n}(\Gamma; r)$ has the product form $P(z) = \prod_{i=1}^n p_i(z_i)$ with $p_i \in \mathcal{P}_i$. For fixed ϕ ,

$$\mathbb{E}_P[\phi(Z)] = \sum_{z \in \Omega_n} \phi(z) \prod_{i=1}^n p_i(z_i)$$

is linear in each p_i with the other factors fixed. A linear functional on a compact finite-dimensional polytope attains its maximum at an extreme point: if a maximizer is not extreme, one of the endpoints in a nontrivial convex decomposition is also a maximizer, and iteration over the finite face dimension reaches an extreme point. Replacing $p_1, \dots, p_n$ successively by extreme-point maximizers never decreases the expectation. The resulting product law lies in $\mathcal{V}_n$, so

$$\mathbb{E}_P[\phi(Z)] \leq \max_{P_v \in \mathcal{V}_n} \mathbb{E}_{P_v}[\phi(Z)].$$

Every $P_v \in \mathcal{V}_n$ is a product law satisfying all bounded-odds inequalities, and hence belongs to $\mathcal{M}_{0,n}(\Gamma; r)$. Taking the supremum over the null class gives the reverse inequality and proves the asserted equality. $\square$

Justification. This lemma converts the bounded-odds product null into the exact finite product-vertex constraints used by every finite primal, dual, validity, and numerical certificate. *Gap.* Without the reduction, the all-tests frontier retains an infinite composite-null supremum and exact size certificates are not finitely auditable. *Consumer.* Conditional sensitivity software can report a maximizing product vertex together with the rejection rule as an exact size certificate.

Theorem 1 (`thm:pair-lf-frontier`). *If $m_i = 2$ for every i , then, under $\text{ass} : \text{sharp} - \text{null}$, $\text{ass} : \text{alternative} - \text{prior} - \text{factorization}$, $\text{ass} : \text{alternative} - \text{record} - \text{factorization}$, $\text{ass} : \text{likelihood} - \text{common} - \text{support}$, $\text{ass} : \text{conditioning} - \text{support}$, and $\text{ass} : \text{causal} - \text{likelihood} - \text{separation}$, $P_n^+ \in \mathcal{M}_{0,n}(\Gamma; r)$, $\phi_{\alpha, P_n^+}^{\text{NP}}(\cdot; r)$ is level α for $\mathcal{M}_{0,n}(\Gamma; r)$, and*

$$\beta_n^*(\alpha; Q, \Gamma, r) = \mathbb{E}_{P_{1,n}^Q}(\cdot | r) [\phi_{\alpha, P_n^+}^{\text{NP}}(Z; r)]$$

for every $\alpha \in (0, 1)$.

Proof. Set $q_i(j) = Q_i(j; r_i) / (Q_i(1; r_i) + Q_i(2; r_i))$. The likelihood common-support and conditioning assumptions, together with $\text{def} : \text{null} - \text{outcome} - \text{law}$, $\text{def} : \text{common} - \text{record} - \text{support}$, and $\text{def} : \text{causal} - \text{bayes} - \text{factor}$, make these probabilities finite and strictly positive. By the two alternative factorization assumptions, the uniform baseline $\text{def} : \text{alternative} - \text{prior}$, and Bayes' formula $\text{def} : \text{alternative} - \text{law}$,

$$q(z) := P_{1,n}^Q(z | r) = \prod_{i=1}^n q_i(z_i).$$

Put $\vartheta_i^+ = \Gamma_i / (1 + \Gamma_i)$. The law P_n^+ in $\text{def} : \text{classical} - \text{tilt}$ is the product law with

$$P_n^+(Z_i = 1) = 1 - \vartheta_i^+ = \frac{1}{1 + \Gamma_i}, \quad P_n^+(Z_i = 2) = \vartheta_i^+ = \frac{\Gamma_i}{1 + \Gamma_i}.$$

Its local probability ratios are Γ_i and Γ_i^{-1} . It therefore satisfies $\text{ass} : \text{conditional} - \text{independence}$ and $\text{ass} : \text{rosenbaum} - \text{odds}$, whence $P_n^+ \in \mathcal{M}_{0,n}(\Gamma; r)$ by $\text{def} : \text{gamma} - \text{null}$.

Write $p^+(z) = P_n^+(z)$. The simple likelihood ratio is

$$L(z) = \frac{q(z)}{p^+(z)} = \prod_{i=1}^n \frac{q_i(z_i)}{p_i^+(z_i)}.$$

Changing coordinate i from position 1 to position 2 multiplies L by

$$\frac{q_i(2)/p_i^+(2)}{q_i(1)/p_i^+(1)} = \frac{Q_i(2; r_i)}{\Gamma_i Q_i(1; r_i)} > 1,$$

where the strict inequality is *ass : causal – likelihood – separation*. Thus L is strictly increasing in the coordinatewise order that codes position 1 as zero and position 2 as one.

Fix $\alpha \in (0, 1)$, as in *ass : test – level*. Finiteness of Ω_n and positivity of p^+ give a likelihood-ratio value $c > 0$ such that

$$P_n^+\{L > c\} \leq \alpha \leq P_n^+\{L \geq c\}.$$

Let

$$\gamma = \frac{\alpha - P_n^+\{L > c\}}{P_n^+\{L = c\}} \in [0, 1], \quad \phi^*(z) = \mathbf{1}\{L(z) > c\} + \gamma \mathbf{1}\{L(z) = c\}.$$

The denominator is positive because c is a value of L . Then $\mathbb{E}_{P_n^+} \phi^* = \alpha$, and ϕ^* is an instance of $\phi_{\alpha, P_n^+}^{\text{NP}}$ in *def : np – rule*, with constant boundary randomization. Strict coordinatewise increase of L also makes ϕ^* coordinatewise nondecreasing, including across its tie set.

For any $P \in \mathcal{M}_{0,n}(\Gamma; r)$, put $\vartheta_i = P(Z_i = 2 \mid r)$. The bounded-odds inequality implies

$$\frac{\vartheta_i}{1 - \vartheta_i} \leq \Gamma_i, \quad \vartheta_i \leq \vartheta_i^+.$$

Conditional independence makes P the product of these Bernoulli-coordinate laws. Couple it to P_n^+ using independent $U_i \sim \text{Unif}(0, 1)$ and the indicators $X_i = \mathbf{1}\{U_i \leq \vartheta_i\}$ and $X_i^+ = \mathbf{1}\{U_i \leq \vartheta_i^+\}$. Then $X_i \leq X_i^+$ almost surely for all i . Coordinatewise monotonicity gives

$$\mathbb{E}_P \phi^* \leq \mathbb{E}_{P_n^+} \phi^* = \alpha.$$

Thus ϕ^* is level α for the full null class.

Finally, let $\psi : \Omega_n \rightarrow [0, 1]$ have $\mathbb{E}_{P_n^+} \psi \leq \alpha$. By the cutoff definition,

$$(\psi(z) - \phi^*(z))(q(z) - cp^+(z)) \leq 0$$

for every z . Therefore

$$\begin{aligned} \mathbb{E}_q(\psi - \phi^*) &= \sum_{z \in \Omega_n} (\psi(z) - \phi^*(z))(q(z) - cp^+(z)) + c\mathbb{E}_{P_n^+}(\psi - \phi^*) \\ &\leq 0. \end{aligned}$$

Every test feasible in *def : frontier* obeys the simple-null size premise because P_n^+ is itself in the composite null, whereas ϕ^* is feasible by the coupling argument. Hence

$$\beta_n^*(\alpha; Q, \Gamma, r) = \mathbb{E}_{P_{1,n}^Q(\cdot \mid r)}[\phi^*(Z)] = \mathbb{E}_{P_{1,n}^Q(\cdot \mid r)}[\phi_{\alpha, P_n^+}^{\text{NP}}(Z; r)].$$

Under *ass : sharp – null*, this conditional assignment calculation is a test of the Fisher sharp causal null at the fixed record r . $\square$

Justification. For pairs, endpoint separation and positivity imply the strict score order needed by the likelihood-ratio argument, so a separate likelihood-order premise is redundant. The classical tilt then dominates every product null for the coordinatewise increasing Neyman–Pearson rule. *Gap.* This is the pair specialization of the conditional all-tests frontier. It does not assert a generic Huber–Strassen representative-pair theorem beyond the directly proved Rosenbaum product argument. *Consumer.* Pair-matched analyses with a prespecified causal alternative may compute the exact conditional likelihood-ratio sensitivity test at every level.

Proposition 1 (*prop:pair-reduction*). *Suppose $m_i = 2$ for every i , and suppose *ass : sharp – null*, *ass : conditional – independence*, *ass : rosenbaum – odds*, *ass : alternative – prior – factorization*, *ass : alternative – record – factorization*, *ass : likelihood – common – support*, *ass : conditioning – support*, and *ass : causal – likelihood – separation* hold. Then $\mathcal{V}_n^Q(r) = \{P_n^+\}$ and $\mathcal{A}_n(Q, \Gamma; r) = (0, 1)$.*

Proof. If $m_i = 2$, the integer $k_i \in \{1, \dots, m_i - 1\}$ in $\text{def} : \text{monotone} - \text{vertices}$ must equal 1. Thus the only terminal block is $S_i = \{2\}$, whose local probabilities are

$$v_{i,\{2\}}(1) = \frac{1}{1 + \Gamma_i}, \quad v_{i,\{2\}}(2) = \frac{\Gamma_i}{1 + \Gamma_i}.$$

These are precisely the probabilities in $\text{def} : \text{classical} - \text{tilt}$, so taking products yields

$$\mathcal{V}_n^Q(r) = \{P_n^+\}.$$

Fix $\alpha \in (0, 1)$. The assumptions of the proposition permit application of $\text{thm} : \text{pair} - \text{lf} - \text{frontier}$, which supplies a generalized Neyman–Pearson rule satisfying

$$\mathbb{E}_{P_n^+}[\phi_{\alpha, P_n^+}^{\text{NP}}(Z; r)] = \alpha, \quad \sup_{P \in \mathcal{M}_{0,n}(\Gamma; r)} \mathbb{E}_P[\phi_{\alpha, P_n^+}^{\text{NP}}(Z; r)] \leq \alpha.$$

Every $P_v \in \mathcal{V}_n$ is a product law and, within each pair, has a local probability ratio in $\{1, \Gamma_i, \Gamma_i^{-1}\}$. Hence $P_v \in \mathcal{M}_{0,n}(\Gamma; r)$, and

$$\max_{P_v \in \mathcal{V}_n} \mathbb{E}_{P_v}[\phi_{\alpha, P_n^+}^{\text{NP}}(Z; r)] \leq \alpha.$$

The cutoff and randomization from the theorem consequently satisfy $\text{def} : \text{validity} - \text{region}$. This proves $(0, 1) \subseteq \mathcal{A}_n(Q, \Gamma; r)$, while the reverse inclusion holds by definition. Therefore

$$\mathcal{A}_n(Q, \Gamma; r) = (0, 1).$$

□

Justification. The pair specialization collapses the indexed terminal-block family to the single classical tilt and makes that tilt minimax sharp at every nondegenerate level. *Gap.* The dependency list records the full causal and conditional testing setup. A separate score-order premise is unnecessary for pairs because positivity and endpoint separation imply $Q_i(2; r_i) > Q_i(1; r_i)$. *Consumer.* The proposition connects the general all-tests formulation to classical pair-matched sign and McNemar sensitivity calibration without introducing a second pair-specific frontier.

Theorem 2 ($\text{thm} : \text{set-mixture-lp-characterization}$). *Under* $\text{ass} : \text{sharp} - \text{null}$, $\text{ass} : \text{conditional} - \text{independence}$, $\text{ass} : \text{rosenbaum} - \text{odds}$, $\text{ass} : \text{alternative} - \text{prior} - \text{factorization}$, $\text{ass} : \text{alternative} - \text{record} - \text{factorization}$, $\text{ass} : \text{likelihood} - \text{common} - \text{support}$, $\text{ass} : \text{conditioning} - \text{support}$, and $\text{ass} : \text{test} - \text{level}$, for $z \in \Omega_n$ and $P_v \in \mathcal{V}_n$, write

$$q_z = P_{1,n}^Q(z \mid r), \quad p_{vz} = P_v(z), \quad \mathcal{V} = \mathcal{V}_n, \quad \mathcal{B} = \mathcal{V}_n^Q(r).$$

The family $\mathcal{B}$ is the indexed terminal-block candidate family; if the scores are additionally strictly increasing as in $\text{ass} : \text{causal} - \text{likelihood} - \text{order}$, its local terminal blocks are upper sets of the Q -ordering. For every nonempty $\mathcal{C} \subseteq \mathcal{V}$, define the computable finite dual value

$$D_{\mathcal{C}}(\alpha) = \min_{y \in [0, \infty)^{\mathcal{C}}, u \in [0, \infty)^{\Omega_n}} \left\{ \alpha \sum_{P_v \in \mathcal{C}} y_v + \sum_{z \in \Omega_n} u_z : \sum_{P_v \in \mathcal{C}} y_v p_{vz} + u_z \geq q_z \quad (z \in \Omega_n) \right\}.$$

Then

$$\beta_n^*(\alpha; Q, \Gamma, r) = D_{\mathcal{V}}(\alpha),$$

and an unrestricted least-favorable convexified mixture over $\mathcal{V}$ exists for every $\alpha \in (0, 1)$. A least-favorable mixture supported on $\mathcal{B}$, with a generalized Neyman–Pearson rule valid for the full Rosenbaum null and attaining β_n^* , exists if and only if

$$D_{\mathcal{B}}(\alpha) = D_{\mathcal{V}}(\alpha). \quad (1)$$

Equivalently, (1) holds if and only if there are $y \in [0, \infty)^{\mathcal{B}} \setminus \{0\}$ and $\rho \in [0, 1]^{\Omega_n}$ such that, with

$$d_y(z) = \sum_{P_v \in \mathcal{B}} y_v p_{vz}, \quad \phi_{y,\rho}(z) = \mathbf{1}\{q_z > d_y(z)\} + \rho(z) \mathbf{1}\{q_z = d_y(z)\},$$

one has the necessary-and-sufficient inequalities and complementary-slackness equalities

$$\sum_z p_{vz} \phi_{y,\rho}(z) \leq \alpha \quad (P_v \in \mathcal{V}), \quad y_v \left(\alpha - \sum_z p_{vz} \phi_{y,\rho}(z) \right) = 0 \quad (P_v \in \mathcal{B}). \quad (2)$$

When (2) holds, $c = \sum_v y_v > 0$, $\lambda(P_v) = y_v/c$, and $d_y = cP_{\lambda}$. The mixture is a dual convexification and need not itself belong to the product null class.

Moreover, $P_n^+ \in \mathcal{V}$, and the exact classical Rosenbaum validity region is

$$\mathcal{A}_n(Q, \Gamma; r) = \{\alpha \in (0, 1) : D_{\{P_n^+\}}(\alpha) = D_{\mathcal{V}}(\alpha)\}. \quad (3)$$

For an explicit interval representation, let

$$\mathcal{T}_+ = \left\{ \frac{q_z}{P_n^+(z)} : z \in \Omega_n \right\}, \quad H_t = \left\{ z : \frac{q_z}{P_n^+(z)} > t \right\}, \quad E_t = \left\{ z : \frac{q_z}{P_n^+(z)} = t \right\},$$

put $h_v(t) = \sum_{z \in H_t} p_{vz}$ and $h_+(t) = \sum_{z \in H_t} P_n^+(z)$, and define

$$K_t = \left\{ \rho \in [0, 1]^{E_t} : h_v(t) + \sum_{z \in E_t} p_{vz} \rho(z) \leq h_+(t) + \sum_{z \in E_t} P_n^+(z) \rho(z) \text{ for every } P_v \in \mathcal{V} \right\}.$$

For nonempty K_t , let a_t and b_t be the minimum and maximum over $\rho \in K_t$ of $h_+(t) + \sum_{z \in E_t} P_n^+(z) \rho(z)$. Then

$$\mathcal{A}_n(Q, \Gamma; r) = (0, 1) \cap \bigcup_{\substack{t \in \mathcal{T}_+ \\ K_t \neq \emptyset}} [a_t, b_t]. \quad (4)$$

Thus $\mathcal{A}_n$ is a finite union of tie-block intervals, and every endpoint is obtained by a finite linear program.

The level-dependent supports are determined by another finite lower envelope. Let $\mathcal{D}_{\mathcal{C}}$ be the polyhedron feasible in the definition of $D_{\mathcal{C}}$. For an extreme point $x = (y^x, u^x)$ of $\mathcal{D}_{\mathcal{C}}$, put $s_x = \sum_v y_v^x$ and $b_x = \sum_z u_z^x$. Then

$$D_{\mathcal{C}}(\alpha) = \min_{x \in \text{ext}(\mathcal{D}_{\mathcal{C}})} \{b_x + \alpha s_x\}. \quad (5)$$

An extreme point $x \in \text{ext}(\mathcal{D}_{\mathcal{B}})$ supplies a full least-favorable indexed support exactly when

$$b_x + \alpha s_x \leq b_{x'} + \alpha s_{x'} \quad \text{for every } x' \in \text{ext}(\mathcal{D}_{\mathcal{V}}). \quad (6)$$

Candidate switch levels are the intersections on the finite lower envelope (5). Between consecutive switch levels the minimizing face is constant; at a switch it is the convex hull of the tied minimizers, and the possible mixture supports are exactly the nonzero y -coordinate supports of points on that face. If

$$\mathcal{I}_{\mathcal{B}} = \{\alpha \in (0, 1) : D_{\mathcal{B}}(\alpha) = D_{\mathcal{V}}(\alpha)\},$$

then every $\mathcal{B}$ -supported least-favorable representation at a level in $\mathcal{I}_{\mathcal{B}} \setminus \mathcal{A}_n(Q, \Gamma; r)$ uses a vertex other than P_n^+ , while outside $\mathcal{I}_{\mathcal{B}}$ no least-favorable mixture supported only on indexed terminal-block candidate vertices exists.

The switch is nonvacuous. For $n = 1$, $m_1 = 3$, $\Gamma_1 = 2$, and

$$(Q_1(1; r_1), Q_1(2; r_1), Q_1(3; r_1)) = (1, 6/5, 3),$$

one has

$$\mathcal{A}_1(Q, 2; r) = (0, 2/3]. \quad (7)$$

At $\alpha = 7/10$, the calibrated classical rule $\phi^+ = (0, 4/5, 1)$ has size $18/25 > 7/10$ at $v_{\{2,3\}} = (1/5, 2/5, 2/5)$. The all-tests optimum and one least-favorable indexed mixture are

$$\phi^* = (1/10, 7/10, 1), \quad \beta_1^*(7/10; Q, 2, r) = 197/260, \quad \lambda = \frac{16}{21}\delta_{v_{\{3\}}} + \frac{5}{21}\delta_{v_{\{2,3\}}}, \quad (8)$$

where $v_{\{3\}} = (1/4, 1/4, 1/2)$ is the classical tilt. Equations (1)–(8) give a finite exact algorithm for the frontier, the validity region, and every level-dependent support switch, while also detecting when the proposed indexed support restriction is invalid.

Proof. By *ass* : likelihood – common – support, *ass* : conditioning – support, and *def* : alternative – law, $q_z > 0$ for every $z \in \Omega_n$, and $\sum_z q_z = 1$. By *lem* : vertex – reduction, the frontier is the finite primal program

$$B(\alpha) = \max_{0 \leq \phi \leq 1} \left\{ \sum_z q_z \phi(z) : \sum_z p_{vz} \phi(z) \leq \alpha \text{ for every } P_v \in \mathcal{V} \right\}. \quad (9)$$

Applying *lem* : finite – lp – strong – duality, with the vertices as rows and assignments as columns, gives the attained dual $B(\alpha) = D_{\mathcal{V}}(\alpha)$. This proves the frontier formula and supplies an optimum on each side.

For any nonempty $\mathcal{C} \subseteq \mathcal{V}$, extending a $D_{\mathcal{C}}$ -feasible vector by zero on $\mathcal{V} \setminus \mathcal{C}$ makes it feasible for $D_{\mathcal{V}}$. Hence

$$D_{\mathcal{C}}(\alpha) \geq D_{\mathcal{V}}(\alpha). \quad (10)$$

For a primal-feasible ϕ and dual-feasible (y, u) , direct expansion gives the nonnegative gap

$$\begin{aligned} & \alpha \sum_v y_v + \sum_z u_z - \sum_z q_z \phi(z) \\ &= \sum_v y_v \left(\alpha - \sum_z p_{vz} \phi(z) \right) + \sum_z u_z (1 - \phi(z)) \\ &+ \sum_z \phi(z) \left(\sum_v y_v p_{vz} + u_z - q_z \right). \end{aligned} \quad (11)$$

Equality holds exactly when every displayed nonnegative product vanishes. Given $y \geq 0$, the smallest coordinatewise slack is $u_z = [q_z - d_y(z)]_+$. The last two complementary-slackness families in (11) then force $\phi(z) = 1$ when $q_z > d_y(z)$, force $\phi(z) = 0$ when $q_z < d_y(z)$, and permit an arbitrary value in $[0, 1]$ when equality holds. The first family is precisely the second condition in (2), while primal feasibility is its first condition. Thus a $\mathcal{B}$ -supported dual certificate attains the full dual value if and only if (1), equivalently (2), holds.

The optimum in (9) is strictly below one. Indeed, since every $q_z > 0$, power one would force $\phi(z) = 1$ for every z , violating every null constraint because $\alpha < 1$. A dual-feasible point with $y = 0$ must have $u_z \geq q_z$, hence objective at least one. Therefore each optimal certificate has $c = \sum_v y_v > 0$. Put $\lambda(P_v) = y_v/c$. Then $d_y = cP_{\lambda}$, and the rule in (2) is the generalized Neyman–Pearson rule against P_{λ} . Complementary slackness gives

$$\mathbb{E}_{P_{\lambda}}[\phi] = \sum_v \lambda(P_v) \mathbb{E}_{P_v}[\phi] = \sum_{v: y_v > 0} \lambda(P_v) \alpha = \alpha. \quad (12)$$

It is calibrated under the convexified mixture, valid at every vertex, valid for the full product null by *lem* : vertex – reduction, and optimal in (9). The mixture is a dual adversary and need not itself be a product member of $\mathcal{M}_{0,n}(\Gamma; r)$. Conversely, suppose a generalized Neyman–Pearson rule against a $\mathcal{B}$ -supported mixture is calibrated under that mixture, valid at every vertex, and optimal in (9). Each support vertex binds, because each expectation is at most α and their mixture average is α . Its cutoff c is positive: cutoff zero would reject at every assignment because $q_z > 0$, contradicting calibration at $\alpha < 1$. Taking $y_v = c\lambda(P_v)$ and $u_z = [q_z - cP_{\lambda}(z)]_+$ makes every term in (11) vanish. Hence it is a $D_{\mathcal{B}}$ -certificate of value $D_{\mathcal{V}}$, proving (1). Taking $\mathcal{B} = \mathcal{V}$ proves unrestricted least-favorable-mixture existence at every level.

Choosing $S_i = \{m_i\}$ in every set shows from *def* : extreme – laws and *def* : classical – tilt that $P_n^+ \in \mathcal{V}$. Applying the support equivalence with $\mathcal{C} = \{P_n^+\}$ gives (3). To verify that this agrees exactly with *def* : validity – region, let ϕ

be an NP rule calibrated under P_n^+ and feasible for the composite null, and let ψ be any other composite-null-feasible rule. Since $P_n^+ \in \mathcal{V}$,

$$\mathbb{E}_{P_n^+}[\psi] \leq \alpha = \mathbb{E}_{P_n^+}[\phi].$$

With likelihood-ratio cutoff t , the NP sign inequality and the preceding size inequality give

$$\sum_z (q_z - tP_n^+(z))(\psi(z) - \phi(z)) \leq 0, \quad t \sum_z P_n^+(z)(\psi(z) - \phi(z)) \leq 0.$$

Adding them yields $\mathbb{E}_q[\psi] \leq \mathbb{E}_q[\phi]$. Thus validity of the calibrated classical rule is sufficient for full optimality, and the converse follows from the singleton dual certificate.

Every generalized NP rule against P_n^+ consists of the strict likelihood-ratio upper set H_t , a tie vector ρ on E_t , and zero rejection below the cutoff. A cutoff strictly between consecutive distinct likelihood-ratio values gives the same rule after moving it to the lower value and assigning zero on the added tie coordinates. Since $0 < \alpha < 1$, it suffices to enumerate $t \in \mathcal{T}_+$. Calibration gives

$$\alpha = h_+(t) + \sum_{z \in E_t} P_n^+(z)\rho(z). \quad (13)$$

Validity at every vertex is exactly the system defining K_t . This polytope is compact and convex, so its image under the scalar affine map in (13) is the closed interval $[a_t, b_t]$. The finite union over cutoffs proves (4), including every boundary point.

The feasible region $\mathcal{D}_C$ is a pointed polyhedron. Its intersection with an objective sublevel set is bounded because all variables are nonnegative and the objective coefficients α and 1 are strictly positive. Hence an optimum is attained at an extreme point. A polyhedron defined by finitely many inequalities has finitely many extreme points, because each is determined by a full-rank selection of active constraints. Evaluating the objective at these points proves (5). A finite lower envelope of affine functions can change its minimizing set only at intersections of lines with distinct slope–intercept pairs. Off those finitely many levels the minimizing face is constant; at a switch the optimal face is the convex hull of the tied minimizers. Extending a point of $\mathcal{D}_B$ by zero outside $\mathcal{B}$ proves (6). Normalizing its nonzero y -coordinates yields the asserted mixture support. Equation (3) then proves the claim on $\mathcal{I}_B \setminus \mathcal{A}_n$, and (1) proves nonexistence of a $\mathcal{B}$ -supported optimum outside $\mathcal{I}_B$.

It remains to verify that a switch actually occurs. The example is compatible with the common-support likelihood assumptions. Take record space $\{r_1, s_1\}$, put $F_{01}(r_1) = 1/4$ and $F_{01}(s_1) = 3/4$, prescribe the likelihood ratios at r_1 as $1, 6/5, 3$, and prescribe them at s_1 as

$$\frac{1 - (1/4)Q_1(j; r_1)}{3/4}, \quad j = 1, 2, 3.$$

The latter values are $1, 14/15, 1/3$; all ratios are finite, strictly positive, and integrate to one. Under the uniform prior,

$$q = (5/26, 6/26, 15/26), \quad v_{\{3\}} = (1/4, 1/4, 1/2), \quad v_{\{2,3\}} = (1/5, 2/5, 2/5). \quad (14)$$

The likelihood ratios under $v_{\{3\}}$ have order $3 > 2 > 1$. For $0 < \alpha \leq 1/2$, the calibrated classical rule is $(0, 0, 2\alpha)$, and it is valid because every local null assigns at most $1/2$ to position three. For $1/2 < \alpha \leq 3/4$, it is $(0, 4\alpha - 2, 1)$. At the vertices indexed by $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$, respectively, its rejection probabilities are

$$\alpha - \frac{1}{4}, \quad 2\alpha - \frac{3}{4}, \quad \alpha, \quad \frac{8\alpha}{5} - \frac{3}{5}, \quad \frac{4\alpha}{5}, \quad \frac{8\alpha - 2}{5}. \quad (15)$$

They are all at most α exactly through $\alpha = 2/3$. For $3/4 < \alpha < 1$, the classical rule is $(4\alpha - 3, 1, 1)$, whose expectation under $v_{\{2,3\}}$ is $4\alpha/5 + 1/5 > \alpha$. Hence (7) holds, and at $\alpha = 7/10$ the classical size at $v_{\{2,3\}}$ is $18/25$.

At that level, take

$$\phi^* = (1/10, 7/10, 1), \quad y_{v_{\{3\}}} = 8/13, \quad y_{v_{\{2,3\}}} = 5/26, \quad u = (0, 0, 5/26). \quad (16)$$

The six vertex expectations are

$$19/40, \quad 5/8, \quad 7/10, \quad 13/25, \quad 29/50, \quad 7/10. \quad (17)$$

Thus ϕ^* is primal feasible and both positively weighted vertices bind. Moreover,

$$d_y = (5/26, 6/26, 10/26),$$

so the dual inequalities and every complementary-slackness equation hold. The common primal and dual value is

$$q^\top \phi^* = 197/260 = \frac{7}{10} \frac{21}{26} + \frac{5}{26}. \quad (18)$$

Both vertices are indexed terminal-block vertices, and normalizing y gives weights 16/21 and 5/21. This proves the strict classical-tilt failure and the asserted least-favorable support switch. $\square$

Lemma 2 (lem:finite-lp-strong-duality). *Let $M, N \geq 1$, let A be a nonnegative M -by- N matrix whose rows are probability vectors, let q be a strictly positive probability vector in $\mathbb{R}^N$, and let $0 < \alpha < 1$. Then*

$$\max_{0 \leq \phi \leq 1} \{q^\top \phi : A\phi \leq \alpha \mathbf{1}\} = \min_{y, u \geq 0} \{\alpha \mathbf{1}^\top y + \mathbf{1}^\top u : A^\top y + u \geq q\},$$

and both optima are attained.

Proof. The primal feasible set is nonempty and compact, so it has a finite attained optimum β . If ϕ is primal feasible and (y, u) is dual feasible, then

$$q^\top \phi \leq (A^\top y + u)^\top \phi \leq y^\top (\alpha \mathbf{1}) + u^\top \mathbf{1}. \quad (1)$$

This proves weak duality.

For the reverse inequality, define

$$D = \{(s, t) \in \mathbb{R}^M \times \mathbb{R} : s = A\phi + h, t = q^\top \phi - k, 0 \leq \phi \leq 1, h \geq 0, k \geq 0\}.$$

This is the sum of a compact convex set and a closed orthant, hence is closed and convex. For $\varepsilon > 0$, the point $x_\varepsilon = (\alpha \mathbf{1}, \beta + \varepsilon)$ is not in D : membership would give some ϕ with $A\phi \leq \alpha \mathbf{1}$ and $q^\top \phi \geq \beta + \varepsilon$, contrary to the definition of β . Let d_ε minimize Euclidean distance from x_ε to D . Existence follows from closedness after intersecting with a sufficiently large closed ball. Differentiating squared distance at zero along the segment from d_ε toward any $d \in D$ gives

$$(x_\varepsilon - d_\varepsilon)^\top (d - d_\varepsilon) \leq 0 \quad (d \in D). \quad (2)$$

Because D is upward closed in its first coordinate and downward closed in its last coordinate, (2) implies that $x_\varepsilon - d_\varepsilon = (-\tilde{y}_\varepsilon, \mu_\varepsilon)$ with $\tilde{y}_\varepsilon \geq 0$ and $\mu_\varepsilon \geq 0$. In fact $\mu_\varepsilon > 0$. Otherwise strict separation of x_ε from D , evaluated at $(0, 0) \in D$, would imply $-\alpha \mathbf{1}^\top \tilde{y}_\varepsilon > 0$, which is impossible. Normalize by μ_ε , writing $y_\varepsilon = \tilde{y}_\varepsilon / \mu_\varepsilon$. Strict separation then yields

$$\beta + \varepsilon > \alpha \mathbf{1}^\top y_\varepsilon + \max_{0 \leq \phi \leq 1} (q - A^\top y_\varepsilon)^\top \phi = \alpha \mathbf{1}^\top y_\varepsilon + \sum_{z=1}^N [q_z - (A^\top y_\varepsilon)_z]_+. \quad (3)$$

Set $u_{\varepsilon, z} = [q_z - (A^\top y_\varepsilon)_z]_+$. Then $(y_\varepsilon, u_\varepsilon)$ is dual feasible and has objective below $\beta + \varepsilon$. For $0 < \varepsilon \leq 1$, (3), nonnegativity, and $\alpha > 0$ bound y_ε and u_ε uniformly. Along a sequence $\varepsilon \downarrow 0$, take a convergent subsequence. Its limit is dual feasible and, by continuity, has objective at most β . Weak duality (1) gives the reverse inequality, so equality holds and the dual optimum is attained. $\square$

Proposition 2 (prop:coordinatewise-monotonicity-loss). *Consider $n = 2$, $(m_1, m_2) = (4, 3)$, $\Gamma_1 = \Gamma_2 = 2$, $\alpha = 1/2$, $Q_1 = (1, 2, 3, 4)$, and $Q_2 = (1, 2, 3)$, so that the conditional alternative mass is $q_{ab} = ab/60$. Call a test coordinatewise monotone when $\phi_{ab} \leq \phi_{a+1, b}$ for $a < 4$ and $\phi_{ab} \leq \phi_{a, b+1}$ for $b < 3$. The unrestricted all-tests optimum is attained by*

$$\phi^{\text{all}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1/2 & 3/4 \\ 0 & 1 & 1 \end{pmatrix}, \quad \beta_{\text{all}} = \frac{143}{240}.$$

The optimum over coordinatewise-monotone tests is attained by

$$\phi^{\text{mon}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1/3 & 5/6 \\ 0 & 5/12 & 5/6 \\ 0 & 11/12 & 1 \end{pmatrix}, \quad \beta_{\text{mon}} = \frac{107}{180}.$$

Consequently, restricting the all-tests Rosenbaum frontier to coordinatewise-monotone tests loses the strict amount $\beta_{\text{all}} - \beta_{\text{mon}} = 1/720$.

Proof. By the conditional alternative formula,

$$q_{ab} = \frac{Q_1(a)}{\sum_{c=1}^4 Q_1(c)} \frac{Q_2(b)}{\sum_{d=1}^3 Q_2(d)} = \frac{a}{10} \frac{b}{6} = \frac{ab}{60}. \quad (1)$$

The numerical scores satisfy strict likelihood ordering and endpoint separation because $4 > 2$ and $3 > 2$. They are also compatible with the common-support likelihood assumptions: for each i , take a two-point record space $\{r_i, s_i\}$, put $F_{0i}(r_i) = 1/8$, prescribe $dF_{1ij}/dF_{0i}(r_i) = Q_i(j)$, and put $dF_{1ij}/dF_{0i}(s_i) = (8 - Q_i(j))/7$. Every displayed density is finite and strictly positive and integrates to one. Product assignment and record laws then realize the stipulated conditional instance.

Let $\mathcal{S}_4$ and $\mathcal{S}_3$ be the nonempty proper subsets of $\{1, 2, 3, 4\}$ and $\{1, 2, 3\}$, respectively. By the extreme-law definition and the vertex-reduction lemma, all $14 \cdot 6 = 84$ null vertices are, without omission,

$$P_{S,T}(a, b) = \frac{2^{\mathbf{1}\{a \in S\}} 2^{\mathbf{1}\{b \in T\}}}{4 + |S| \ 3 + |T|}, \quad (S, T) \in \mathcal{S}_4 \times \mathcal{S}_3. \quad (2)$$

For a matrix $\phi = (\phi_{ab})$, write $P_{S,T}\phi = \sum_{a=1}^4 \sum_{b=1}^3 P_{S,T}(a, b)\phi_{ab}$. Direct substitution of the two proposed rules into all vertices in (2) gives the following exact row maxima over the six choices of T . In the row order

$$S = (1), (2), (12), (3), (13), (23), (123), (4), (14), (24), (124), (34), (134), (234), \quad (3)$$

the values for ϕ^{all} are

$$\left(\frac{7}{20}, \frac{9}{20}, \frac{3}{8}, \frac{9}{20}, \frac{3}{8}, \frac{11}{24}, \frac{11}{28}, \frac{1}{2}, \frac{5}{12}, \frac{1}{2}, \frac{3}{7}, \frac{1}{2}, \frac{3}{7}, \frac{1}{2} \right), \quad (4)$$

and those for ϕ^{mon} are

$$\left(\frac{7}{20}, \frac{9}{20}, \frac{3}{8}, \frac{109}{240}, \frac{109}{288}, \frac{133}{288}, \frac{19}{48}, \frac{1}{2}, \frac{5}{12}, \frac{143}{288}, \frac{143}{336}, \frac{1}{2}, \frac{3}{7}, \frac{1}{2} \right). \quad (5)$$

Every entry in (4) and (5) is at most $1/2$, so both rules satisfy all 84 null inequalities. Their entries lie in $[0, 1]$, and the second matrix visibly satisfies every horizontal and vertical order inequality. Equation (1) gives their powers

$$\sum_{a,b} q_{ab} \phi_{ab}^{\text{all}} = \frac{143}{240}, \quad \sum_{a,b} q_{ab} \phi_{ab}^{\text{mon}} = \frac{107}{180}. \quad (6)$$

It remains to certify that neither feasible value can be improved. Let A be the 84-by-12 matrix with rows $P_{S,T}$, and consider first the unrestricted primal. Its dual has nonnegative variables $y_{S,T}$ for the null constraints, u_{ab} for $\phi_{ab} \leq 1$, and ℓ_{ab} for $-\phi_{ab} \leq 0$, with stationarity and objective

$$A^\top y + u - \ell = q, \quad \frac{1}{2} \sum_{S,T} y_{S,T} + \sum_{a,b} u_{ab}. \quad (7)$$

Set

$$y_{4,3} = \frac{1}{3}, \quad y_{4,23} = \frac{5}{24}, \quad y_{34,3} = \frac{2}{5}, \quad y_{34,23} = \frac{1}{4}, \quad (8)$$

with all other y 's and every u equal to zero, and set

$$\ell = \begin{pmatrix} 1/30 & 1/30 & 1/20 \\ 1/60 & 0 & 0 \\ 1/40 & 0 & 0 \\ 1/30 & 0 & 0 \end{pmatrix}. \quad (9)$$

Multiplication using (2) gives $A^\top y - \ell = q$, so (7) holds coordinate by coordinate and all dual variables are nonnegative. The four constraints indexed by $(4, 3), (4, 23), (34, 3), (34, 23)$ each have expectation $1/2$ under ϕ^{all} . Every positive entry of ℓ corresponds to a zero entry of ϕ^{all} , while $u = 0$. Thus all complementary-slackness products vanish. The dual value is

$$\frac{1}{2} \left(\frac{1}{3} + \frac{5}{24} + \frac{2}{5} + \frac{1}{4} \right) = \frac{143}{240}. \quad (10)$$

Weak duality, or equivalently the finite strong-duality lemma, and (6) prove the unrestricted optimum.

For the monotone primal, orient each covering order inequality from a lower cell l to an adjacent higher cell h as $\phi_l - \phi_h \leq 0$, and give it multiplier $w_{l \rightarrow h} \geq 0$. The stationarity equation becomes

$$\sum_{S,T} y_{S,T} P_{S,T}(a,b) + u_{ab} + \sum_{h:(a,b) \rightarrow h} w_{(a,b) \rightarrow h} - \sum_{l:l \rightarrow (a,b)} w_{l \rightarrow (a,b)} - \ell_{ab} = q_{ab}. \quad (11)$$

Take the only nonzero variables, apart from ℓ , to be

$$y_{4,23} = \frac{5}{12}, \quad y_{34,3} = \frac{2}{3}, \quad y_{34,23} = \frac{1}{12}, \quad w_{(2,3) \rightarrow (3,3)} = \frac{1}{180}, \quad u_{4,3} = \frac{1}{90}, \quad (12)$$

and set

$$\ell = \begin{pmatrix} 11/360 & 1/30 & 2/45 \\ 1/72 & 0 & 0 \\ 1/36 & 0 & 0 \\ 1/36 & 0 & 0 \end{pmatrix}. \quad (13)$$

Substitution of (2) into (11) gives equality to the matrix $q = (ab/60)$ in every coordinate; equivalently, the residual before subtracting ℓ is exactly $q + \ell$. The three positively weighted null constraints $(4, 23), (34, 3), (34, 23)$ bind at $1/2$. The weighted order constraint binds because $\phi_{2,3}^{\text{mon}} = \phi_{3,3}^{\text{mon}} = 5/6$; the weighted upper-box constraint binds because $\phi_{4,3}^{\text{mon}} = 1$; and every positive lower-box multiplier in (13) is paired with a zero entry of ϕ^{mon} . Hence primal feasibility, dual feasibility, and every complementary-slackness equation hold. The monotone dual objective is

$$\frac{1}{2} \left(\frac{5}{12} + \frac{2}{3} + \frac{1}{12} \right) + \frac{1}{90} = \frac{107}{180}. \quad (14)$$

This equals the monotone primal value in (6), proving optimality. Finally,

$$\frac{143}{240} - \frac{107}{180} = \frac{429 - 428}{720} = \frac{1}{720} > 0,$$

which proves the strict loss. $\square$

Lemma 3 (lem:bounded-independent-hoeffding). *Let $X_1, \dots, X_n$ be independent real random variables satisfying $a_i \leq X_i \leq b_i$ almost surely, where $a_i < b_i$. Then, for every $t > 0$,*

$$\Pr \left(\sum_{i=1}^n (X_i - \mathbb{E} X_i) \geq t \right) \leq \exp \left\{ - \frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right\},$$

and

$$\Pr \left(\sum_{i=1}^n (X_i - \mathbb{E} X_i) \leq -t \right) \leq \exp \left\{ - \frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right\}.$$

Proof. Put $V = \sum_{i=1}^n (b_i - a_i)^2$. We first prove the bounded moment-generating-function inequality. Let $X \in [a, b]$ almost surely, let $\mu = \mathbb{E}X$, put $p = (\mu - a)/(b - a) \in [0, 1]$, and let $h = \lambda(b - a)$. Convexity of $x \mapsto e^{\lambda x}$ gives

$$\mathbb{E}e^{\lambda(X-\mu)} \leq (1-p)e^{-ph} + pe^{(1-p)h}. \quad (1)$$

For $g(h) = \log(1-p+pe^h) - ph$, direct differentiation yields $g(0) = g'(0) = 0$ and

$$g''(h) = \frac{p(1-p)e^h}{(1-p+pe^h)^2} = w_h(1-w_h) \leq \frac{1}{4}, \quad w_h = \frac{pe^h}{1-p+pe^h}. \quad (2)$$

Integrating (2) twice from zero to h , with the integration direction reversed when $h < 0$, gives $g(h) \leq h^2/8$. Thus (1) implies

$$\mathbb{E}e^{\lambda(X-\mu)} \leq \exp\left\{\frac{\lambda^2(b-a)^2}{8}\right\} \quad (\lambda \in \mathbb{R}). \quad (3)$$

By independence and (3), for every $\lambda > 0$,

$$\mathbb{E} \exp\left\{\lambda \sum_{i=1}^n (X_i - \mathbb{E}X_i)\right\} \leq \exp\left\{\frac{\lambda^2 V}{8}\right\}. \quad (4)$$

Markov's inequality therefore gives

$$\Pr\left(\sum_{i=1}^n (X_i - \mathbb{E}X_i) \geq t\right) \leq \exp\left\{-\lambda t + \frac{\lambda^2 V}{8}\right\}. \quad (5)$$

The choice $\lambda = 4t/V$ turns the exponent in (5) into $-2t^2/V$, proving the upper-tail inequality. Applying that result to $-X_1, \dots, -X_n$, whose range lengths remain $b_i - a_i$, proves the lower-tail inequality. $\square$

Justification. The lemma provides the uniform concentration step for independent but potentially non-identical null factors in the replicated experiment. *Gap.* The null factors may vary by set, so the repeated-design proof needs a bound uniform over the entire product class rather than an iid-only tail calculation. *Consumer.* The replicated theorem applies both tails with score range length $\Gamma + 1$.

Lemma 4 (lem:explicit-rosenbaum-endpoint-separator). *Under* $\text{ass} : \text{replicated} - \text{conditional} - \text{design}$ *and* $\text{def} : \text{local} - \text{rosenbaum} - \text{polytope}$, *fix* $1 \leq \Gamma < Q(m)/Q(1)$ *and define*

$$s_\Gamma(j) = \mathbf{1}\{j = m\} - \Gamma \mathbf{1}\{j = 1\}.$$

Then

$$\sup_{p \in \mathcal{P}_m(\Gamma)} p^\top s_\Gamma = 0,$$

with equality at the local vertex whose unnormalized weights are Γ at position m and 1 at every other position. For $q_j = Q(j)/\sum_{k=1}^m Q(k)$,

$$\Delta(Q, \Gamma) := q^\top s_\Gamma = \frac{Q(m) - \Gamma Q(1)}{\sum_{k=1}^m Q(k)} > 0.$$

Proof. For any $p \in \mathcal{P}_m(\Gamma)$, the division-free Rosenbaum inequality in $\text{def} : \text{local} - \text{rosenbaum} - \text{polytope}$, applied with $j = m$ and $k = 1$, gives $p_m \leq \Gamma p_1$. Hence

$$p^\top s_\Gamma = p_m - \Gamma p_1 \leq 0. \quad (1)$$

Let p^+ have normalized weights Γ at position m and 1 at every other position, so

$$p_m^+ = \frac{\Gamma}{\Gamma + m - 1}, \quad p_j^+ = \frac{1}{\Gamma + m - 1} \quad (j \neq m). \quad (2)$$

All coordinates in (2) are positive, sum to one, and the ratio of any two is at most Γ ; thus $p^+ \in \mathcal{P}_m(\Gamma)$. Moreover, $p_m^+ - \Gamma p_1^+ = 0$. Together with (1), this proves both the supremum formula and its attainment.

Under $\text{ass} : \text{replicated} - \text{conditional} - \text{design}$, $q_j = Q(j)/\sum_{k=1}^m Q(k)$. Therefore

$$q^\top s_\Gamma = q_m - \Gamma q_1 = \frac{Q(m) - \Gamma Q(1)}{\sum_{k=1}^m Q(k)}. \quad (3)$$

The denominator is strictly positive because $Q \in (0, \infty)^m$, and the numerator is strictly positive because $\Gamma < Q(m)/Q(1)$. Equation (3) is the asserted positive margin. $\square$

Justification. The explicit endpoint score supplies the concrete separating hyperplane required for the repeated-design achievability proof. *Gap.* An abstract existence witness would not provide the oracle test or the exact achievable exponential constant used downstream. *Consumer.* The repeated-design theorem uses its zero null support value and positive alternative margin directly.

Theorem 3 (thm:replicated-all-tests-design-sensitivity). *Under* $\text{ass} : \text{replicated} - \text{conditional} - \text{design}$ *and* $\text{ass} : \text{test} - \text{level}$,

$$\tilde{\Gamma}_{\text{all}}^*(Q) = \frac{Q(m)}{Q(1)}.$$

More precisely, if $1 \leq \Gamma < Q(m)/Q(1)$, let s_Γ and $\Delta(Q, \Gamma)$ be as in lem : explicit – rosenbaum – endpoint – separator, and define

$$c_0(Q, \Gamma) = \frac{\Delta(Q, \Gamma)^2}{2(\Gamma + 1)^2}, \quad \phi_n(z) = \mathbf{1} \left\{ \frac{1}{n} \sum_{i=1}^n s_\Gamma(z_i) > \frac{\Delta(Q, \Gamma)}{2} \right\}.$$

Then

$$\sup_{p_1, \dots, p_n \in \mathcal{P}_m(\Gamma)} \mathbb{E}_{\otimes_{i=1}^n p_i} [\phi_n(Z)] \leq e^{-c_0(Q, \Gamma)n}, \quad \mathbb{E}_{q^{\otimes n}} [1 - \phi_n(Z)] \leq e^{-c_0(Q, \Gamma)n}.$$

With

$$N_0(\alpha, Q, \Gamma) = \left\lceil \frac{\log(1/\alpha)}{c_0(Q, \Gamma)} \right\rceil,$$

every $n \geq N_0(\alpha, Q, \Gamma)$ satisfies

$$\beta_{n, \text{rep}}^*(\alpha; Q, \Gamma) \geq 1 - e^{-c_0(Q, \Gamma)n}.$$

If $\Gamma \geq Q(m)/Q(1)$, then $\beta_{n, \text{rep}}^*(\alpha; Q, \Gamma) = \alpha$ for every n . When $m = 2$, the boundary obeys

$$\frac{Q(2)}{Q(1)} = \frac{q_2}{1 - q_2},$$

the sign/McNemar success odds.

Proof. Fix $1 \leq \Gamma < Q(m)/Q(1)$. By lem : explicit – rosenbaum – endpoint – separator, every $p_i \in \mathcal{P}_m(\Gamma)$ satisfies

$$\mathbb{E}_{p_i} [s_\Gamma(Z_i)] \leq 0, \quad \mathbb{E}_q [s_\Gamma(Z_i)] = \Delta(Q, \Gamma) > 0. \quad (1)$$

The score takes only the values $-\Gamma, 0, 1$, so its range length is $\Gamma + 1$. Under an arbitrary null product $\otimes_{i=1}^n p_i$, the variables $X_i = s_\Gamma(Z_i)$ are independent, possibly non-identically distributed, and the rejection event implies

$$\sum_{i=1}^n (X_i - \mathbb{E}_{p_i} X_i) > \frac{n\Delta(Q, \Gamma)}{2}, \quad (2)$$

because $\sum_i \mathbb{E}_{p_i} X_i \leq 0$. Applying the upper-tail inequality in lem : bounded – independent – hoeffding with $t = n\Delta(Q, \Gamma)/2$ and $b_i - a_i = \Gamma + 1$ gives

$$\mathbb{E}_{\otimes_{i=1}^n p_i} [\phi_n(Z)] \leq \exp \left\{ -\frac{2(n\Delta/2)^2}{n(\Gamma + 1)^2} \right\} = e^{-c_0(Q, \Gamma)n}. \quad (3)$$

Since the right side is independent of the factors, taking their supremum proves the first asserted bound.

Under the alternative $q^{\otimes n}$, the X_i are independent with common mean $\Delta(Q, \Gamma)$. The event $1 - \phi_n = 1$ implies

$$\sum_{i=1}^n (X_i - \Delta(Q, \Gamma)) \leq -\frac{n\Delta(Q, \Gamma)}{2}. \quad (4)$$

The lower-tail inequality in lem : bounded – independent – hoeffding yields the second bound $e^{-c_0(Q, \Gamma)n}$. If $n \geq N_0(\alpha, Q, \Gamma)$, then $e^{-c_0n} \leq \alpha$. Thus ϕ_n is feasible in def : replicated – frontier, and (4) gives

$$\beta_{n, \text{rep}}^*(\alpha; Q, \Gamma) \geq \mathbb{E}_{q^{\otimes n}} \phi_n \geq 1 - e^{-c_0(Q, \Gamma)n}. \quad (5)$$

In particular, the frontier tends to one for every $\Gamma < Q(m)/Q(1)$.

Now suppose $\Gamma \geq Q(m)/Q(1)$. The endpoint-extremality condition in *ass : replicated – conditional – design* implies, for every j, k ,

$$\frac{q_j}{q_k} = \frac{Q(j)}{Q(k)} \leq \frac{Q(m)}{Q(1)} \leq \Gamma. \quad (6)$$

Thus $q \in \mathcal{P}_m(\Gamma)$, and $q^{\otimes n}$ is one of the null product laws in *def : replicated – frontier*. Every feasible test consequently has alternative power at most α . The constant randomized test $\phi(z) = \alpha$ has expectation exactly α under every law, so it is feasible and attains that upper bound. Therefore

$$\beta_{n,\text{rep}}^*(\alpha; Q, \Gamma) = \alpha \quad (n \geq 1). \quad (7)$$

Because $Q(m) > Q(1)$ and $0 < \alpha < 1$, (5)–(7) show that the set in *def : all – tests – design – sensitivity* contains every $\Gamma \in [1, Q(m)/Q(1))$ and no $\Gamma \geq Q(m)/Q(1)$. Its supremum is therefore $Q(m)/Q(1)$. Finally, if $m = 2$, then

$$q_2 = \frac{Q(2)}{Q(1) + Q(2)}, \quad 1 - q_2 = \frac{Q(1)}{Q(1) + Q(2)},$$

so $Q(2)/Q(1) = q_2/(1 - q_2)$, the success odds in the sign/McNemar reduction. No optimal exponential-rate claim is used or implied. $\square$

Justification. This theorem identifies the maximal common hidden-bias level below which an explicit randomized conditional test has power tending to one, supplies finite- n exponential error bounds, and proves the exact all-tests converse at and above the boundary. *Gap.* Published design-sensitivity calculations generally fix a statistic; they do not establish this all-tests oracle ceiling together with an explicit endpoint witness and exact converse. *Consumer.* For pairs it recovers the sign/McNemar odds boundary, while for larger fixed sets it benchmarks what any practical statistic could possibly achieve when Q is known.

7 Interpretation

The causal target is the Fisher sharp null, whereas $\beta_n^*(\alpha; Q, \Gamma, r)$ is the conditional all-tests power frontier after fixing the record. Positive causal Bayes factors determine the conditional alternative assignment law, while bounded Rosenbaum odds determine the composite product-law null. A least-favorable mixture is a convexified dual certificate and generally is not itself an independent assignment law in $\mathcal{M}_{0,n}(\Gamma; r)$. The indexed terminal-block family requires no likelihood-order assumption; it becomes a Q -upper-set family only when strict score ordering is separately imposed. In the replicated benchmark the separating score is explicit: $s_\Gamma(j) = \mathbf{1}\{j = m\} - \Gamma \mathbf{1}\{j = 1\}$. It is an oracle, Q -known endpoint score, not the classical tilt, a selected practical rank statistic, a coherent combination, or an estimator when Q is unknown.

Technical internal limitation (diagnostic only)

The finite characterization is exact, but the product-vertex count can grow exponentially. Indexed terminal-block support is certified by $D_{\mathcal{B}} = D_{\mathcal{V}}$, not guaranteed. The $n = 2$, sizes $(4, 3)$, common- $\Gamma = 2$ diagnostic checks all 84 vertices and exact unrestricted and monotone duals, proving optima 143/240 and 107/180 and gap 1/720; it is an obstruction to an unsafe monotonicity restriction, not a separate field-level novelty claim. Endpoint-touching components such as $(0, 2/3]$ are intersections of ambiently closed projections with $(0, 1)$ and need not be closed in $\mathbb{R}$. The replicated theorem uses the explicit endpoint score and gives the achievable constant $c_0 = \Delta^2/[2(\Gamma + 1)^2]$; it does not claim the optimal Chernoff exponent.

8 Honest scope

The finite results are conditional on the sharp-null sensitivity model, the factorized uniform-prior causal alternative, common likelihood support, and the fixed observed record. They deliver the exact all-tests envelope, product-vertex validity certificates, indexed-support criterion, validity-region topology, affine support diagnostics, the three-position switch, and the exact monotonicity-loss diagnostic. The asymptotic result is only for fixed m , fixed record, fixed endpoint-extremal Q , common Γ , and independent conditional replicas whose null factors may vary across sets. It

does not optimize the exponent, handle heterogeneous or growing sets, estimate Q , prove unconditional validity, construct a selected rank statistic or coherent combination, establish the proposed two-sided 0.158 claim, prove Bonferroni equivalence, or identify a statistic-choice optimum.

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